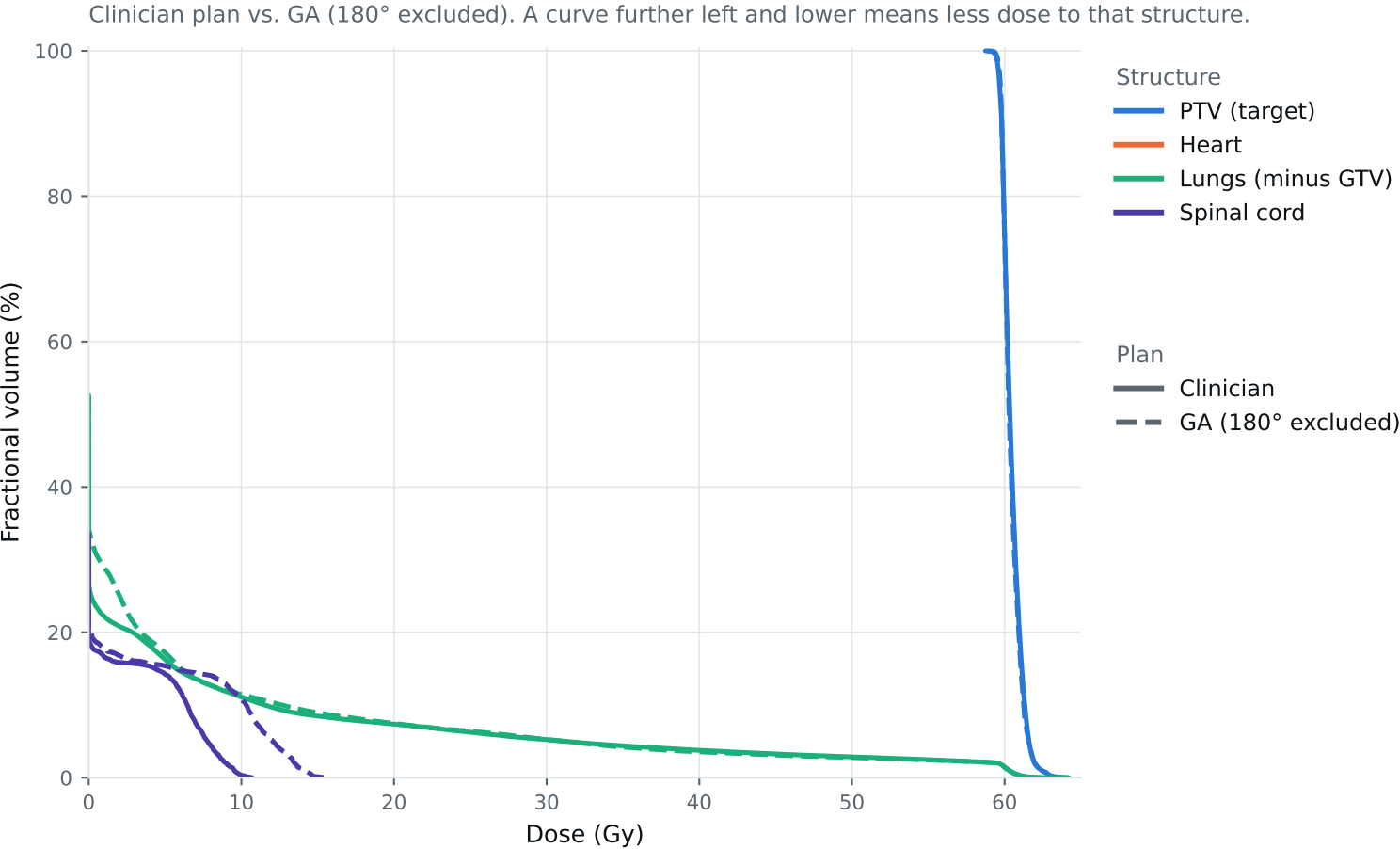


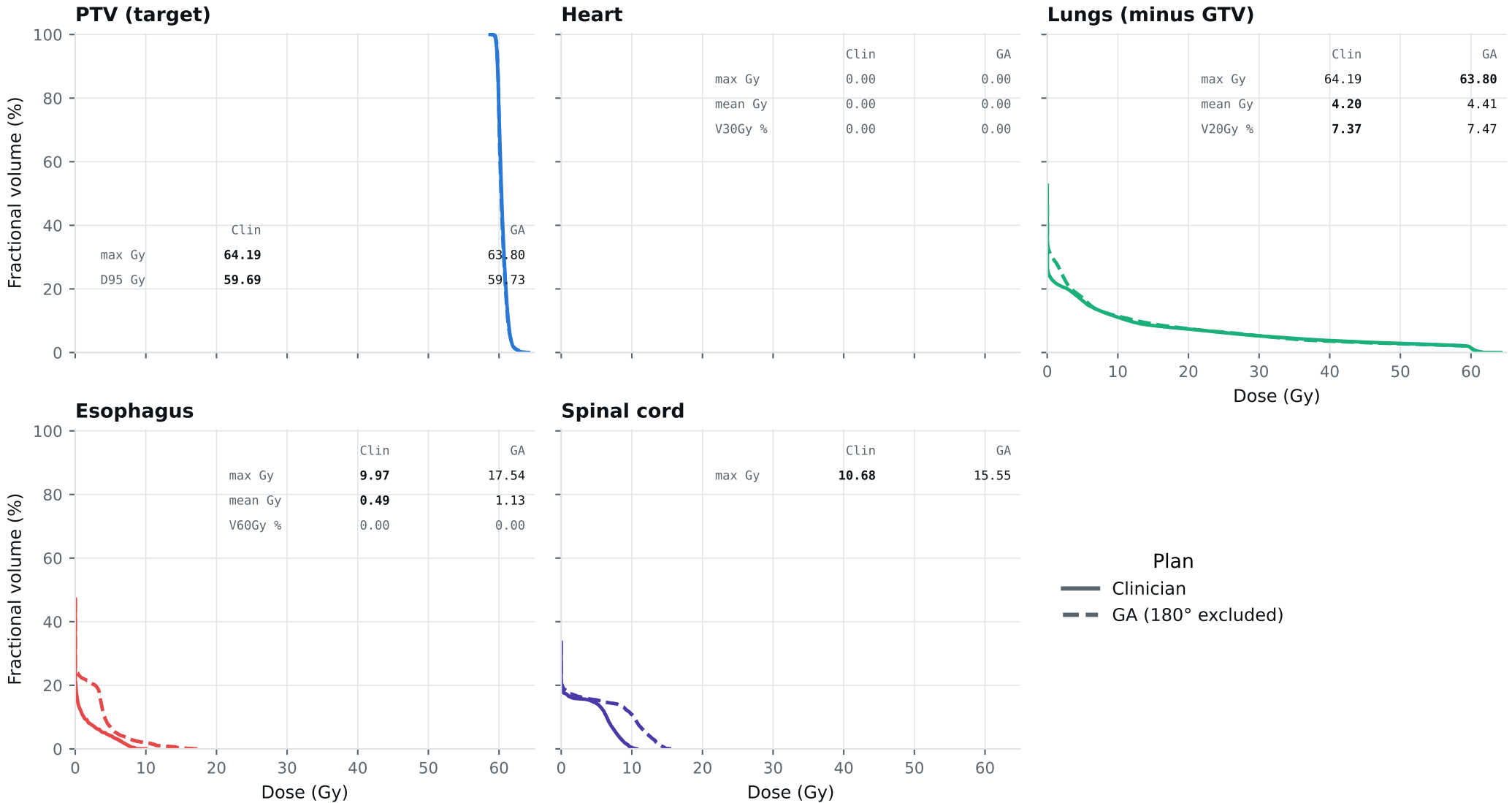
# Dose-volume histogram — Lung\_Patient\_37



Curves further left are better for organs; the PTV curve should stay high, then drop sharply. Target curves should overlap; organ curves show the tradeoffs. Esophagus is on page 2.

# DVH by structure — Lung\_Patient\_37

Same curves, one structure per panel, all plans, each annotated with that structure's protocol metrics.  
Bold = clinically preferable: organs favor lower dose; PTV favors the value closest to goal within the limit. Grey = all plans are within 0.01 of each other. Amber misses a goal,



# Clinical criteria — Lung\_Patient\_37

Organs: lower is better. PTV: closest to goal without exceeding the limit. Bold marks the best plan.

| Criterion                       | Limit     | Goal                                 | Clinician    | GA<br>180° excluded |    |
|---------------------------------|-----------|--------------------------------------|--------------|---------------------|----|
| GTV max                         | 69        | 66                                   | 63.55        | <b>63.36</b>        | Gy |
| PTV max                         | 69        | 66                                   | <b>64.19</b> | 63.80               | Gy |
| ESOPHAGUS max                   | 66        | —                                    | <b>9.97</b>  | 17.54               | Gy |
| ESOPHAGUS mean                  | 34        | 21                                   | <b>0.49</b>  | 1.13                | Gy |
| ESOPHAGUS V60Gy                 | 17        | —                                    | 0.00         | 0.00                | %  |
| HEART max                       | 66        | —                                    | 0.00         | 0.00                | Gy |
| HEART mean                      | 27        | 20                                   | 0.00         | 0.00                | Gy |
| HEART V30Gy                     | 50        | 48                                   | 0.00         | 0.00                | %  |
| LUNG_L max                      | 66        | —                                    | 64.19        | <b>63.80</b>        | Gy |
| LUNG_R max                      | 66        | —                                    | <b>9.45</b>  | 15.71               | Gy |
| CORD max                        | 50        | 48                                   | <b>10.68</b> | 15.55               | Gy |
| SKIN max                        | 60        | —                                    | <b>45.35</b> | 45.73               | Gy |
| LUNGS_NOT_GTV max               | 66        | —                                    | 64.19        | <b>63.80</b>        | Gy |
| LUNGS_NOT_GTV mean              | 21        | 20                                   | <b>4.20</b>  | 4.41                | Gy |
| LUNGS_NOT_GTV V20Gy             | 37        | —                                    | <b>7.37</b>  | 7.47                | %  |
| PTV D95 (target coverage)       |           |                                      | 59.69        | 59.73               |    |
| Objective value (search signal) |           |                                      | 45.59        | 44.96               |    |
| MOSEK solve time                |           |                                      | 13 s         | 13 s                |    |
| Beam angles (gantry°)           | Clinician | 175°, 150°, 125°, 58°, 30°, 0°, 200° |              |                     |    |
|                                 | GA        | 0°, 60°, 90°, 135°, 165°, 195°, 225° |              |                     |    |

Grey rows: all plans agree within 0.01 — heart, lung and LUNGS\_NOT\_GTV max sit at 66 Gy because those structures abut the target. Amber misses the goal; red exceeds the limit.

Source: Lung\_Patient\_37\_clinical\_metrics\_full\_resolution.json, full resolution. The curves and table come from the same solve.